Rabiul Awal

Department of Computer Science and Operations Research
University of Montreal, QC, Canada
Mila - Quebec AI Institute, QC, Canada

Email: awalrabiul6@gmail.com
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Research Interests: Visual Representation Learning, Generative Models, World Models

Current Position

09/2024 – present **Mila - Quebec AI Institute**
Graduate Researcher

Education

09/2024 – present **University of Montreal, Mila - Quebec AI Institute**
Ph.D. in Computer Science
Department of Computer Science and Operations Research
Advisor: Aishwarya Agrawal
Courses: Reinforcement Learning and Optimal Control, Vision and Language

01/2021 – 12/2022 **University of Saskatchewan**
MSc. Computer Science
Advisor: Michael Horsch, Roy Ka-Wei Lee

03/2014 – 12/2018 **Noakhali Science and Technology University**
BSc. Computer Science
Noakhali, Bangladesh

Previous Experience

07/2024 – 08/2025 **ServiceNow Research**
Visiting Researcher in the Multimodal Team with Sai Rajeswar

01/2023 – 08/2024 **Mila - Quebec AI Institute**
Research Intern with Aishwarya Agrawal *Quebec, Canada*

03/2022 – 09/2022 **Mitacs, LST Accelerate Program**
Machine Learning Intern *Saskatoon, Canada*

06/2020 – 11/2022 **Singapore University of Technology and Design**
Student Researcher at Social AI Studio with Roy Ka-Wei Lee *Remote*

01/2019 – 12/2020 **Goava Sales Intelligence AB**
Software Engineer *Dhaka, Bangladesh*

Selected Awards, Grant & Fellowship

2025 **Outstanding Reviewer at CVPR**

2024 **Selected for CIFAR DLRL Summer School at Toronto**

2024 **Selected for Upper Bound Talent Bursary at Alberta ($1500)**

2022 **Selected for Alberta University AI Week Talent Bursary ($1500)**

2022 **D&I Awardee at ACL Virtual Conference**

2021 **Graduate Teaching Fellowship at University of Saskatchewan, Canada**

2016 **1st place at NSTU Intra University Programming Contest, Dhaka, Bangladesh**

2015 **3rd place at CoU Regional Programming Contest (CoURPC), Comilla, Bangladesh**

Peer-Reviewed Conference Publications

Google Scholar: `https://scholar.google.com/citations?user=oOcLfUMAAAAJ&hl=en`
*Denotes equal contribution

[*C9*] Juan A. Rodriguez, Haotian Zhang, Abhay Puri, Aarash Feizi, Rishav Pramanik, Pascal Wichmann, Arnab Mondal, Mohammad Reza Samsami, **Rabiul Awal** + 7 others. Rendering-Aware Reinforcement Learning for Vector Graphics Generation. *Conference on Neural Information Processing Systems* (*NeurIPS*). 2025.

[*C8*] Saba Ahmadi*, **Rabiul Awal***, Ankur Sikarwar*, Amirhossein Kazemnejad*, Ge Ya Luo, Juan A Rodriguez, Sai Rajeswar, Siva Reddy, Christopher Pal, Benno Krojer, Aishwarya Agrawal. The Promise of RL for Autoregressive Image Editing. *Conference on Neural Information Processing Systems* (*NeurIPS*). 2025.
Code and models available at `https://github.com/mair-lab/EARL`.

[*C7*] Anikter Didolkar*, Andrii Zadaianchuk*, **Rabiul Awal***, Maximilian Seitzer, Efstratios Gavves, Aishwarya Agrawal. CTRL-O: Language-Controllable Object-Centric Visual Representation Learning. *IEEE / CVF Computer Vision and Pattern Recognition Conference* (*CVPR*). 2025.
Spotlight presentation at Multimodal Algorithmic Reasoning Workshop, CVPR'25.

[*C6*] **Rabiul Awal**, Mahsa Massoud, Aarash Feizi, Zichao Li + 9 others. WebMMU: A Benchmark for Multimodal Multilingual Website Understanding and Code Generation. *The Conference on Empirical Methods in Natural Language Processing* (*EMNLP*). 2025.
Spotlight presentation at Multimodal Algorithmic Reasoning Workshop, CVPR.

[*C5*] **Rabiul Awal***, Saba Ahmadi*, Le Zhang*, and Aishwarya Agrawal. VisMin: Visual Minimal-Change Understanding. *The Conference on Neural Information Processing Systems* (*NeurIPS*). 2024.
Contributed talk at SpLU-RoboNLP Workshop, ACL 2024.

[*C4*] Shravan Nayak, Kanishk Jain, **Rabiul Awal**, Siva Reddy, Sjoerd van Steenkiste, Lisa Anne Hendricks, Karolina Stanczak, Aishwarya Agrawal. Benchmarking Vision Language Models for Cultural Understanding. *The Conference on Empirical Methods in Natural Language Processing* (*EMNLP*). 2024.
Oral Presentation.

[*C3*] Le Zhang, **Rabiul Awal**, Aishwarya Agrawal. Contrasting Intra-modal and Ranking Cross-modal Hard Negatives to Enhance Visio-linguistic Compositional Understanding. *IEEE / CVF Computer Vision and Pattern Recognition Conference* (*CVPR*). 2024.
Spotlight presentation at the Open-Domain Reasoning Multi-Modal Workshop, CVPR'23.

[*C2*] Han Wang, Ming Shan Hee, **Rabiul Awal**, Kenny Tsu Wei Choo, Roy Ka-Wei Lee. Evaluating GPT-3 Generated Explanations for Hateful Content Moderation. *International Joint Conference on Artificial Intelligence* (*IJCAI*). 2023.

[*C1*] **Rabiul Awal**, Roy Ka-Wei Lee, Eshaan Tanwar, Tanmay Garg, Tanmoy Chakraborty. Model-agnostic meta-learning for multilingual hate speech detection. *IEEE Transactions on Computational Social Systems*. 2023.

[*C0*] **Rabiul Awal**, Rui Cao, Roy Ka-Wei Lee, Sandra Mitrović. AngryBERT: Joint learning target and emotion for hate speech detection. *Pacific-Asia Conference on Knowledge Discovery and Data Mining* (*PAKDD*). 2021.

Service

Broader Community

2025–present **Whiteboard Session Lead Organizer**
Help organize the RL for Foundation Models reading group for internal lab at Mila

2021–2022 **Computer Science Graduate Council MSc Representative**
Helped in strengthening graduate research activities, hosting social events for grad students at University of Saskatchewan

2023–2024 **Language Grounding Reading Group Organizer**
Help supporting running the reading group on vision language at Mila

2021–2022 **Computer Science Graduate Council MSc Representative**
Helped in strengthening graduate research activities, hosting social events for grad students at University of Saskatchewan

2021–2023 **NLPDhaka Reading Group Founder**
A grass-root initiative for fostering ML/NLP research in Bangladesh
Organized ∼35 crowd-source reading sessions to spread AI research in local community

2017–2018 **NSTU Computing Club President**
Led the computing club growth 10X (from 15 to 150 members), conducted training on competitive programming

Research Community

02/2026 **Workshop Organizer at Mila**
World Modeling Workshop

05/2025 **Workshop Organizer at CVPR25**
Vision Language Models For All: Building Geo-Diverse and Culturally Aware Vision-Language Models

2021–present **Reviewer**
AAAI26, NeurIPS25, CVPR25, ECCV24, CVPR24, ACL23, SIGIR22, AACL22, ICDM21, SIGKDD21

2021 **Volunteer**
ICML21, EACL21, ICLR21, NAACL21

Teaching

09/2025 – 12/2025 **Mila – Quebec AI Institute, Canada**
Teaching Assistant
Computer Science Department
Courses: Representation Learning

01/2021 – 12/2022 **University of Saskatchewan, Canada**
Graduate Teaching Assistant
Computer Science Department
Courses: Data Structure and Algorithms, Python, Introduction to Artificial Intelligence

06/2017 – 06/2018 **Noakhali Science and Technology University, Bangladesh**
Teaching Assistant
CSTE 3110 Microprocessor, Micro-controller and Interfacing Lab

In-progress Research

The Promise of Reward Modeling for Image Synthesis: Reward modeling to align diffusion models with human aesthetic and semantic preferences through efficient RL training.
Latent Few-step Diffusion World Model: Designing a more accurate and long-horizon few-step world model that enables efficient policy training with improved representation learning.

Skills

Python (PyTorch, Tensorflow), Java, C/C++, NoSQL (Elasticsearch), AWS (SageMaker, EC2, Lambda)